Notes

Introduction to Calculus and Functions

Overview of the Session

This session will focus on building foundational knowledge for calculus. The primary goal is to prepare students for understanding functions, which is a crucial prerequisite for calculus.

Importance of Functions

Calculus heavily relies on a strong understanding of functions. Before diving into calculus, it is essential to master concepts related to functions.

Prerequisites for Functions

To effectively learn functions, students should first be familiar with the concepts of sets and relations. This session will begin by covering sets, then move on to functions, and finally, progress to calculus.

Session Approach

- Basic concepts will be covered quickly.
- More complex or typical topics will be given ample time for detailed explanation and discussion.
- Active participation and quick responses from students are encouraged to ensure an efficient and smooth learning process.

Introduction and Session Guidelines

Importance of Notes

- **Handwritten notes are crucial:** There is no substitute for handwritten notes. Students should make their own notes during the session.
- **Integrating existing notes:** If students have existing notes from 11th and 12th grade, they should either create separate new notes or continuously add new pointers and highlight repetitive information in their existing notes.
- **Conciseness:** Notes should be short and crisp, focusing on key information rather than detailed explanations.

Session Expectations

- **Active Participation:** Students are expected to be fully aligned and ready with a copy and pen.

Practice and Illustrations: Substantial theory will be followed by practice questions and in-depth illustrations as part of the daily routine.

Introduction to Sets

What are Sets?

Sets are fundamental to mathematics, especially in calculus, and are defined as a **collection of well-defined objects, articles, or elements**.

Representation of Sets

- **Notation:** Sets are typically represented using **capital letters** (e.g., A, B, C).
- **Elements:** The individual items within a set are called elements. These are usually represented by **small alphabets** (e.g., a, b, c).
- **Curly Brackets:** Elements of a set are enclosed within curly brackets and separated by commas. For example, a set A containing elements a, b, and c would be written as $A = \{a, b, c\}$.

Belonging to a Set

- The symbol "∈" is used to denote that an element "belongs to" a set. For instance, if 'a' is an element of set A, we write $a \in A$.

Repetition of Elements

- In set theory, the repetition of elements within a set is not considered. If an element appears multiple times, it is still counted as a single distinct element. For example, the set $\{a, b, c, a\}$ is considered the same as $\{a, b, c\}$. It is conventional to write each distinct element only once.

Representing a Set: Roster Form and Set-Builder Form

Two Ways to Represent a Set

There are two primary methods to represent a set:

1. **Listing all elements (Roster Form)**: In this method, all the elements of the set are explicitly written down, separated by commas, and enclosed within curly braces.
2. **Stating a property (Set-Builder Form)**: In this method, a common property that all elements of the set satisfy is described. This property uniquely defines the elements belonging to the set.

Roster Form

- **Definition**: This form involves listing every element of the set.

Notes

Set-Builder Form

- **Definition**: This form describes the elements of a set by stating a property that all its elements satisfy.
- **Notation**: It is typically written as $\{x \mid \text{property of } x\}$ or $\{x : \text{property of } x\}$, where 'x' represents an element of the set, and the colon or vertical bar means "such that".
- **Example**: For the set of vowels, instead of listing them, we can describe the property: $A = \{x \mid x \text{ is a vowel}\}$. This means "A is the set of all x such that x is a vowel."

Matching Roster Form with Set-Builder Notation

Understanding Set Representation

This section focuses on matching sets represented in roster form with their corresponding descriptions in set-builder notation. We will analyze examples to understand how to identify the elements of a set based on given conditions.

Example 1: Divisors of Six

Consider the set given in roster form: $1, 2, 3, 6$.

We need to find a description that matches these elements. Let's analyze the condition "X is a prime divisor of six."

1. **Divisors of six:** The numbers that divide six evenly are $1, 2, 3, 6$.
2. **Prime divisors of six:** Among these divisors, the prime numbers are 2 and 3.

Therefore, the set described by "X is a prime divisor of six" is $2, 3$, which does not match the given set $1, 2, 3, 6$.

Example 2: Letters of the Word "Mathematics"

Consider the set of letters: A, C, E, H, I, M, S, T.

Let's check if this set matches the description "X is a letter of the word 'mathematics'."

1. **Letters in "mathematics":** M, A, T, H, E, M, A, T, I, C, S.
2. **Unique letters in "mathematics":** A, C, E, H, I, M, S, T.

By comparing, we can see that the set A, C, E, H, I, M, S, T perfectly matches the unique letters present in the word "mathematics".

Conclusion

Set-Builder Notation Example

Example: Representing a Set in Set-Builder Notation

Let's consider a set with the following elements: $A = \{1, 2, 4, 8, 16, 32, 64\}$. We need to represent this set using set-builder notation.

Observation of Property

Upon observing the elements, we can identify a pattern:

- $1 = 2^0$
- $2 = 2^1$
- $4 = 2^2$
- $8 = 2^3$
- $16 = 2^4$
- $32 = 2^5$
- $64 = 2^6$

This indicates that each element in the set is a power of 2.

Method 1: Using Whole Numbers

One way to represent this set is by defining the elements as 2^n, where n belongs to the set of whole numbers. Since the highest power is 6, we must specify the upper limit for n.

$$A = \{x \mid x = 2^n, \text{ where } n \in \mathbb{W} \text{ and } n \leqslant 6\}$$

Here, $\mathbb{W}$ denotes the set of whole numbers ($\{0, 1, 2, 3, \ldots\}$). If we substitute $n = 0, 1, 2, 3, 4, 5, 6$, we get all the elements of set A.

Method 2: Using Natural Numbers

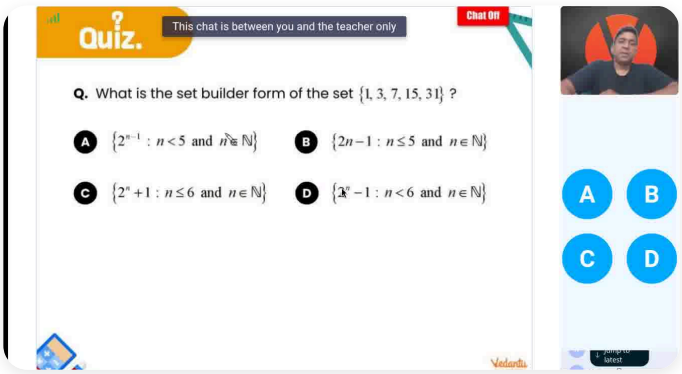
Another way to represent the set is by defining the elements as 2^{n-1}, where n belongs to the set of natural numbers. In this case, the exponent starts from $n - 1 = 0$ when $n = 1$, and goes up to $n - 1 = 6$ when $n = 7$. Therefore, n must be less than or equal to 7.

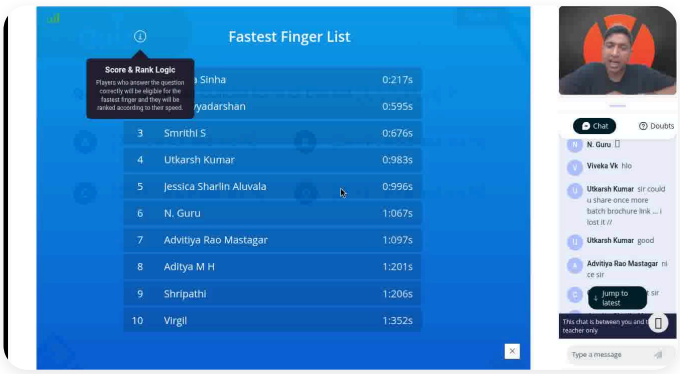
$$B = \{x \mid x = 2^{n-1}, \text{ where } n \in \mathbb{N} \text{ and } n \leqslant 7\}$$

Here, $\mathbb{N}$ denotes the set of natural numbers ($\{1, 2, 3, \ldots\}$). If we substitute $n = 1, 2, 3, 4, 5, 6, 7$, we get all the elements of set A.

The choice of representation depends on how one perceives the pattern. As long as the notation accurately describes all elements of the set and excludes any other elements, it is correct. When you write down all the elements using either definition, you will obtain the same original set.

Quiz: Set Builder Form Identification





Question

What is the set builder form of the set $\{1, 3, 7, 15, 31\}$?

Options: A. $\{2^{n-1} : n < 5 \text{ and } n \in N\}$ B. $\{2n - 1 : n \leqslant 5 \text{ and } n \in N\}$ C. $\{2^n + 1 : n \leqslant 6 \text{ and } n \in N\}$ D. $\{2^n - 1 : n < 6 \text{ and } n \in N\}$

Solution Explanation

To find the set builder form for the given set $\{1, 3, 7, 15, 31\}$, we need to identify a pattern that generates these numbers.

Let's examine the elements:

- $1 = 2^1 - 1$
- $3 = 2^2 - 1$
- $7 = 2^3 - 1$
- $15 = 2^4 - 1$
- $31 = 2^5 - 1$

From this pattern, it is clear that each element can be represented in the form $2^n - 1$.

Now, we need to determine the range of 'n'. Since the first element corresponds to $n = 1$ and the last element corresponds to $n = 5$, 'n' must be a natural number ($n \in N$) and its value must be from 1 to 5, inclusive.

This can be expressed as $n \leqslant 5$ and $n \in N$. Alternatively, it can be expressed as $n < 6$ and $n \in N$, as natural numbers less than 6 are $1, 2, 3, 4, 5$.

Notes

Fastest Finger List

The following students answered correctly and were ranked by speed:

1. Sinha
2. Priyadarshan
3. Smrithi S
4. Utkarsh Kumar
5. Jessica Sharlin Aluvala
6. N. Guru
7. Advitya Rao Mastagar
8. Aditya M H
9. Shripathi
10. Virgil

Special Varieties of Sets: Null Set

Definition of a Null Set

A **Null Set**, also known as an **Empty Set** or **Void Set**, is a set that contains no elements. It is denoted by the symbol ϕ (phi) or by an empty pair of curly brackets, `{}` .

Identifying a Null Set

While the concept of a null set is straightforward, problems often present conditions that implicitly define a null set. You need to analyze the given conditions to determine if any elements can satisfy them. If no elements can satisfy the conditions, the set is a null set.

Examples of Null Sets

1. **Example 1:** Consider a set A defined as: $A = \{x \mid x^2 + 1 = 0, x \in \mathbb{R}\}$

   To determine if A is a null set, we need to find if there is any real number x such that $x^2 + 1 = 0$.

   - If x is a real number, then $x^2 \geq 0$.
   - Therefore, $x^2 + 1 \geq 1$.
   - This means $x^2 + 1$ can never be equal to 0 for any real number x.

   Thus, set A contains no elements, making it a null set. $A = \phi$.

- From the equation $2x - 3 = 0$, we get $2x = 3$, which implies $x = \frac{3}{2}$.
- The condition states that x must be an integer ($x \in \mathbb{Z}$). However, $\frac{3}{2}$ is not an integer.

Thus, set B contains no elements that satisfy both conditions, making it a null set. $B = \phi$.

Singleton Set

Definition

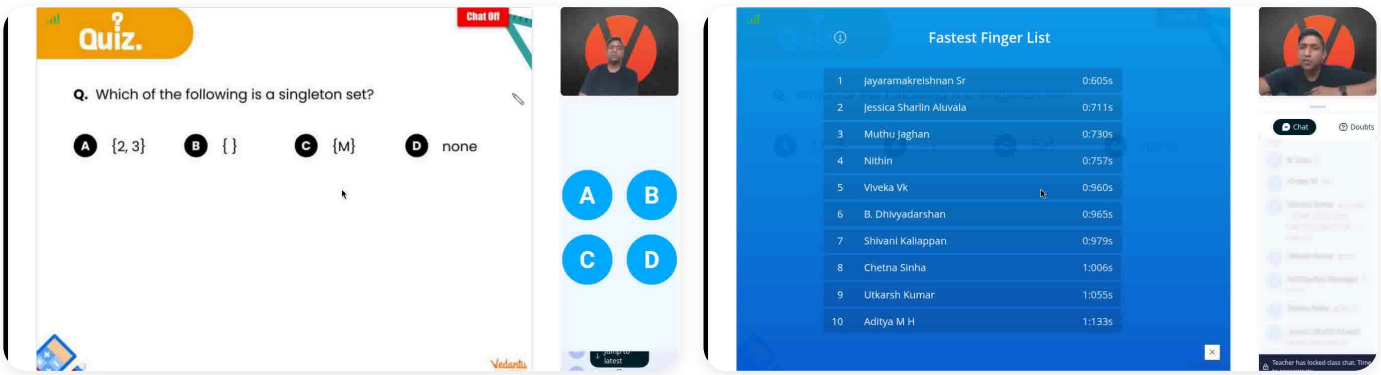
A singleton set is a set that contains exactly one element. It is also known as a unit set.

Examples

- If a set contains only one element, it is a singleton set.
- Representations of a singleton set:
  - {a}
  - {1}
  - {x}

In simple terms, if you have a curly bracket with just one entry inside, it is a singleton set.

Quiz: Identifying a Singleton Set



Question

Which of the following is a singleton set?

A. {2, 3} B. {} C. {M} D. none

Explanation

To identify a singleton set, we need to understand its definition. A singleton set is a set that contains exactly one element.

- **Option B: {}** This set contains no elements. It is an empty set or a null set. Therefore, it is not a singleton set.
- **Option C: {M}** This set contains exactly one element, which is 'M'. Therefore, this is a singleton set.

Conclusion

The correct answer is **C. {M}** because it is the only set among the options that contains exactly one element.

Finite Set

Definition of a Finite Set

A finite set is a set that contains a finite number of elements. This means that the elements in the set can be counted, and the counting process will eventually come to an end.

Example

- **Is a set of natural numbers an example of a finite set?**
  - **Answer:** False.
  - **Explanation:** The set of natural numbers (1, 2, 3, ...) is an infinite set because there is no end to the number of elements it contains. Therefore, it is not a finite set.

Key Takeaway

If the number of elements in a set is finite, then and only then is it called a finite set. If the number of elements is infinite, it is an infinite set.

Cardinality of a Set

Definition of Cardinality

The cardinality of a set refers to the number of elements present in that set. It is a fundamental concept in set theory, indicating the "size" of a set.

Notation

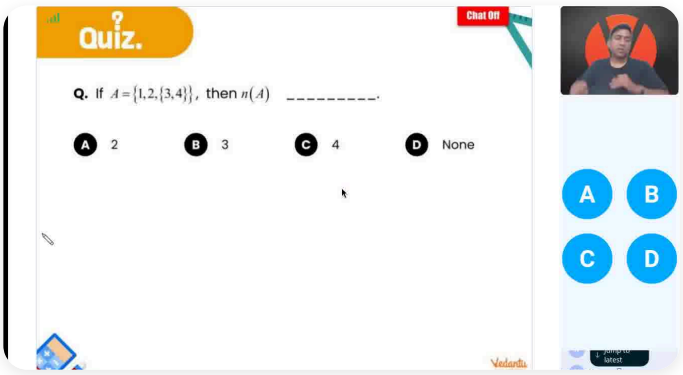
The cardinality of a set A is commonly denoted as $N(A)$ or $|A|$. Both notations represent the same concept: the count of distinct elements within the set.

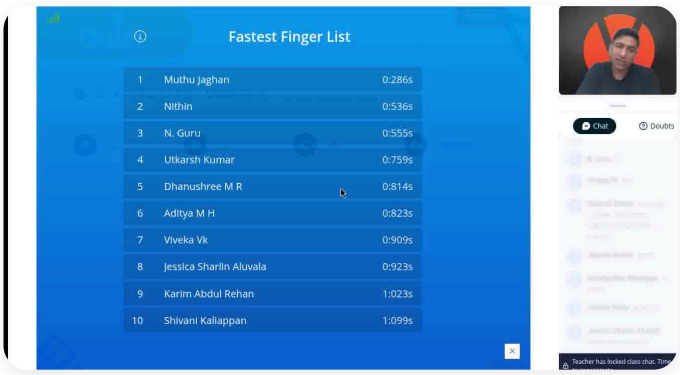
Example

Consider a set A given by A = {1, 2, 3, 6, 7}.

Therefore, the cardinality of set A is 5, which can be written as $N(A) = 5$.

Quiz: Number of Elements in a Set





Question

If $A = \{1, 2, \{3, 4\}\}$, then $n(A)$ _____ A 2 B 3 C 4 D None

Solution Explanation

Many students get confused when a curly bracket itself represents an element within a set. In the given set $A = \{1, 2, \{3, 4\}\}$, the elements are:

1. The number 1
2. The number 2
3. The set $\{3, 4\}$ (this entire set is considered a single element)

Therefore, the total number of elements in set A, denoted as $n(A)$, is 3.

Correct Answer

B 3

Types of Sets: Null, Singleton, Finite, and Infinite

Review of Set Types

We have previously discussed several types of sets:

- **Null Set:** A set containing no elements.
- **Singleton Set:** A set containing exactly one element.
- **Finite Set:** A set with a countable number of elements. The process of counting the elements in a finite set would eventually come to an end.

the elements, the counting process would never end.

Examples of Infinite Sets

Many common mathematical sets are examples of infinite sets:

- **Set of Integers:** $\{..., -2, -1, 0, 1, 2, ...\}$
- **Set of Natural Numbers:** $\{1, 2, 3, ...\}$
- **Set of Whole Numbers:** $\{0, 1, 2, 3, ...\}$
- **Set of Composite Numbers:** $\{4, 6, 8, 9, 10, ...\}$ (numbers that have more than two factors)

All these sets contain an unending number of elements, hence they are classified as infinite sets.

Difference Between Equal Sets and Equivalent Sets

Introduction

This section clarifies the distinction between equal sets and equivalent sets, which is a fundamental concept in set theory.

Equivalent Sets

Equivalent sets are defined by having the same number of elements, meaning their cardinalities are equal. The specific elements within the sets do not need to be identical.

- **Definition:** Two sets, A and B, are equivalent if $n(A) = n(B)$, where $n(A)$ denotes the cardinality (number of elements) of set A.
- **Example:**
  - Set A = {1, 2, 3}
  - Set B = {dog, cat, parrot}
  - Here, $n(A) = 3$ and $n(B) = 3$. Since their cardinalities are equal, Set A and Set B are equivalent sets.

Equal Sets

Equal sets are defined by containing exactly the same elements. This implies that their cardinalities must also be equal, but the primary condition is the identity of elements.

- **Definition:** Two sets, A and B, are equal if every element of A is an element of B, and every element of B is an element of A. This can be written as $A \subseteq B$ and $B \subseteq A$.
- **Example:**
  - Set A = {1, 2, 3}

- In this case, A = B = C because they all contain the exact same elements, regardless of the order.

Key Difference Summary

- **Equivalent Sets:** Same number of elements (same cardinality).
- **Equal Sets:** Exactly the same elements (implies same cardinality and identical elements).

Introduction to Subsets

Definition of a Subset

A set A is said to be a subset of a set B if every element of A is also an element of B. This means that if an element x belongs to set A, then x must also belong to set B.

Notation

The standard symbol for a subset is $\subseteq$. If A is a subset of B, we write it as $A \subseteq B$.

Example

Let's consider two sets:

- Set B = {1, 2, 3, 4}
- Set A = {1, 2}

In this example, every element of Set A (which are 1 and 2) is also present in Set B. Therefore, we can say that A is a subset of B, written as $A \subseteq B$.

Understanding Subsets

Definition of a Subset

A subset is a set formed by taking some or all of the elements from another set. Every element of a subset must also be an element of the original set.

Listing All Subsets of a Given Set

Let's consider a set A with three elements, for example, $A = \{1, 2, 3\}$. To write down all possible subsets of set A, we consider sets with varying numbers of elements:

- **Subsets with zero elements:** The empty set, denoted by $\varnothing$ or {}, is a subset of every set. For set A, this is {}.
- **Subsets with one element:** These are sets containing each element individually: {1}, {2}, {3}.

Subsets with three elements: This is the set itself: {1, 2, 3}.

Combining all these, the complete list of subsets for A = {1, 2, 3} is:
{∅, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3}, {1, 2, 3}}

Formula for the Total Number of Subsets

To find the total number of subsets for any given set, we use a simple formula. If a set A has 'n' elements, the total number of subsets is given by 2^n.

- **Formula:** Total number of subsets = 2^n where 'n' is the number of elements in the set.
- **Example:** For the set A = {1, 2, 3}, the number of elements 'n' is 3. Therefore, the total number of subsets is 2^3 = 8. This matches our manual listing of subsets above.

This formula is very useful and should be remembered for quick calculations regarding subsets.

Properties of Subsets: Null Set and the Set Itself

Key Properties of Subsets

When considering subsets of any given set, two particular subsets are always present and are considered very important:

- **The Null Set (∅):** The null set, also known as the empty set, is a subset of every set. This is a fundamental property in set theory. It contains no elements, and thus, all its elements (of which there are none) are contained within any other set.
- **The Set Itself:** Every set is a subset of itself. This means that if you have a set A, then A is always a subset of A. This is because all elements of A are, by definition, contained within A.

These two subsets are always included when listing all possible subsets of any given set.

Equality of Sets based on Subsets

Condition for Equality of Sets

If set A is a subset of set B ($A \subseteq B$) and set B is also a subset of set A ($B \subseteq A$), then it implies that set A and set B are equal ($A = B$).

This is a fundamental property in set theory, meaning that if every element of A is in B, and every element of B is in A, then A and B must contain exactly the same elements, making them equal sets.

Example

Consider the statement: A = {1, 2, 3}.

Notes

Introduction to Power Set

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Example 2: Power Set of a Set with Two Elements

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Number of Elements in a Power Set

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Quiz: Number of Elements in Power Set

Quiz

Q. If $n(A) = 3$, then $n(P(A))$ = _____

A. 6 B. 3 C. 8 D. None

Solution

To find the number of elements in the power set of A, denoted as $n(P(A))$, when the number of elements in set A, $n(A)$, is given, we use the formula:

$n(P(A)) = 2^{n(A)}$

Given $n(A) = 3$, we substitute this value into the formula:

$n(P(A)) = 8$

Therefore, the number of elements in the power set of A is 8.

Correct Answer: C. 8

This concept is fundamental for understanding sets and their properties. The total number of subsets of a set is equal to the number of elements in its power set.

Problem: Elements in Power Sets

Problem Statement

Two finite sets have P and Q number of elements respectively. The number of elements in the power set of the first set is 56 more than the number of elements in the power set of the second set. We need to find the combination of P and Q.

Given Data

- Let Set A have P elements, so $n(A) = P$.
- Let Set B have Q elements, so $n(B) = Q$.

Power Set Properties

- The number of elements in the power set of Set A, denoted as $n(P(A))$, is 2^P.
- The number of elements in the power set of Set B, denoted as $n(P(B))$, is 2^Q.

Forming the Equation

According to the problem statement, the number of elements in the power set of the first set is 56 more than the number of elements in the power set of the second set. This can be written as: $n(P(A)) = n(P(B)) + 56$ Substituting the power set formulas: $2^P = 2^Q + 56$ Rearranging the equation: $2^P - 2^Q = 56$

Solving the Equation

To solve for P and Q, we can factor out 2^Q from the left side: $2^Q(2^{P-Q} - 1) = 56$ Now, we need to express 56 as a product of a power of 2 and an odd number (or a number of the form $2^x - 1$). We know that $56 = 8 \times 7$. We can write 8 as 2^3. We can write 7 as $2^3 - 1$. So, the equation becomes: $2^Q(2^{P-Q} - 1) = 2^3(2^3 - 1)$

Comparing Factors

2. Comparing the terms in the parentheses: $2^{P-Q} - 1 = 2^3 - 1 \implies 2^{P-Q} = 2^3 \implies P - Q = 3$

Finding P and Q

We have two equations:

1. $Q = 3$
2. $P - Q = 3$

Substitute the value of Q from the first equation into the second equation: $P - 3 = 3$ $P = 3 + 3$ $P = 6$

Conclusion

The combination of P and Q is $P = 6$ and $Q = 3$.

Quiz: Subsets of Finite Sets

Problem Statement

Two finite sets have m and n number of elements respectively. If the number of subsets of the first set is 112 more than the number of subsets of the second set, find the values of m and n.

Solution

Let the first set have m elements and the second set have n elements.

The number of subsets of a set with k elements is given by 2^k.

According to the problem statement: Number of subsets of the first set = 2^m Number of subsets of the second set = 2^n

We are given that the number of subsets of the first set is 112 more than the number of subsets of the second set. This can be written as: $2^m = 2^n + 112$ Rearranging the equation, we get: $2^m - 2^n = 112$

To solve this equation, we can factor out 2^n from the left side: $2^n(2^{m-n} - 1) = 112$

Now, we need to find factors of 112 that can be expressed in the form 2^n and $(2^{m-n} - 1)$.

Let's find the prime factorization of 112: $112 = 2 \times 56$ $112 = 2 \times 2 \times 28$ $112 = 2 \times 2 \times 2 \times 14$ $112 = 2 \times 2 \times 2 \times 2 \times 7$ So, $112 = 2^4 \times 7$

Now, we can equate the factors: $2^n(2^{m-n} - 1) = 2^4 \times 7$

By comparing the terms, we can deduce:

From the second equation: $2^{m-n} = 7 + 1$ $2^{m-n} = 8$ $2^{m-n} = 2^3$

Comparing the exponents, we get: $m - n = 3$

We already found $n = 4$. Substitute this value into the equation: $m - 4 = 3$ $m = 3 + 4$ $m = 7$

Thus, the values are $m = 7$ and $n = 4$.

Final Answer

$m = 7$ $n = 4$

Operations on Sets

Introduction to Set Operations

In set theory, several fundamental operations allow us to combine or modify sets. These operations are crucial for understanding relationships between sets and solving problems involving collections of elements.

Key Operations

- **Union (∪):** The union of two sets, say A and B, denoted as $A \cup B$, is the set containing all elements that are in A, or in B, or in both. Mathematically, $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$.
- **Intersection (∩):** The intersection of two sets, A and B, denoted as $A \cap B$, is the set containing all elements that are common to both A and B. Mathematically, $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$.
- **Difference (− or ∖):** The difference of two sets, A and B, denoted as $A - B$ or $A \setminus B$, is the set containing all elements that are in A but not in B. Mathematically, $A - B = \{x \mid x \in A \text{ and } x \notin B\}$.
- **Complement (A' or A^c):** The complement of a set A, denoted as A' or A^c, with respect to a universal set U, is the set of all elements in U that are not in A. Mathematically, $A' = \{x \mid x \in U \text{ and } x \notin A\}$.
- **Symmetric Difference (Δ):** The symmetric difference of two sets, A and B, denoted as $A \Delta B$, is the set of elements which are in either of the sets, but not in their intersection. Mathematically, $A \Delta B = (A - B) \cup (B - A) = (A \cup B) - (A \cap B)$.

These operations form the basis for more complex set theory concepts and are frequently used in various fields of mathematics and computer science.

Universal Set

Definition

Examples

- When discussing natural numbers, the set of all natural numbers can be considered the universal set.
- When working with the real number system, the set of all real numbers serves as the universal set.

Representation

In Venn diagrams, the universal set is typically represented by a rectangle, while other sets (like set A, set B) are drawn as circles or ovals within this rectangle.

Union of Sets

Definition of Union

The union of two sets, A and B, denoted as $A \cup B$, is the set containing all elements that are in A, or in B, or in both. It combines all unique elements from both sets.

Example

Let's consider two sets:

- Set A = {1, 2, 3}
- Set B = {4, 5, 6}

To find the union of A and B ($A \cup B$), we list all the elements present in either set A or set B. So, $A \cup B$ will be {1, 2, 3, 4, 5, 6}.

Symbolic Representation

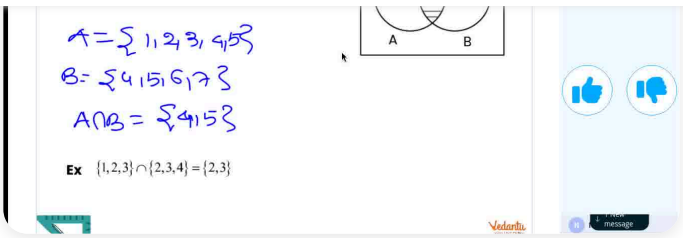
If x is an element of the union of sets A and B ($x \in A \cup B$), it means that x belongs to set A OR x belongs to set B. This can be written as:

$$x \in A \cup B \iff x \in A \text{ or } x \in B$$

Notation

The symbol used to denote the union of sets is $\cup$.

Intersection of Two Sets



Definition of Intersection

Intersection of two sets, denoted as $A \cap B$, consists of elements that are common to both set A and set B. Mathematically, it is defined as:

$$A \cap B = \{x : x \in A \text{ and } x \in B \text{ simultaneously}\}$$

This means an element x belongs to the intersection of A and B if and only if x is an element of A AND x is an element of B.

Example of Intersection

Let's consider two sets:

- Set A = {1, 2, 3, 4, 5}
- Set B = {4, 5, 6, 7}

To find the intersection $A \cap B$, we look for elements that are present in both sets. In this case, the common elements are 4 and 5.

Therefore, $A \cap B$ = {4, 5}.

Another example: {1, 2, 3} ∩ {2, 3, 4} = {2, 3}

Venn Diagram Representation

In a Venn diagram, the intersection of two sets A and B is represented by the common region where the circles for A and B overlap. This common region visually depicts the elements that belong to both sets simultaneously. Understanding this visual representation is crucial for solving many problems involving set operations.

Distributive Law of Set Operations

Understanding the Distributive Law

We are familiar with the concepts of union and intersection of sets. Let's explore a fundamental property related to these operations, known as the Distributive Law.

Consider the expression: $A \cap (B \cup C)$.

1. First, identify the region representing $B \cup C$ (all elements in B or C or both).
2. Then, find the common elements between set A and the region $B \cup C$. This is $A \cap (B \cup C)$.

Now, consider another expression: $(A \cap B) \cup (A \cap C)$.

This expression represents the union of two intersections:

1. First, identify the region representing $A \cap B$ (elements common to A and B).
2. Next, identify the region representing $A \cap C$ (elements common to A and C).
3. Finally, take the union of these two regions, $(A \cap B) \cup (A \cap C)$.

Is $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ True?

By thinking about the Venn diagrams for both expressions, we can conclude that the statement is **true**. This is a fundamental property in set theory, similar to the distributive property in algebra (e.g., $a \times (b + c) = (a \times b) + (a \times c)$).

This property is known as the **Distributive Law of Intersection over Union**.

Distributive Law of Union over Intersection

Similarly, the distributive law also holds for union over intersection:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

This means that the union of set A with the intersection of sets B and C is equal to the intersection of the union of A and B with the union of A and C. This can also be verified using Venn diagrams.

Subset Properties: Union and Intersection

Understanding Union and Intersection when A is a Subset of B

Let's consider a scenario where set A is a subset of set B. This means that every element of A is also an element of B.

To visualize this, imagine a Venn diagram:

- Set A is represented by a smaller circle completely contained within a larger circle representing Set B.

A. Union of A and B ($A \cup B$)

When A is a subset of B, the union of A and B ($A \cup B$) will be the larger set, which is B itself.

B. Intersection of A and B ($A \cap B$)

When A is a subset of B, the intersection of A and B ($A \cap B$) will be the smaller set, which is A itself.

- **Reasoning:** The intersection includes only the elements that are common to both A and B. Since A is entirely contained within B, the elements common to both are precisely the elements of A.
- **Result:** $A \cap B = A$

Disjoint Sets

Definition of Disjoint Sets

Two sets, A and B, are considered **disjoint sets** if their intersection is a null set. This means they have no elements in common.

Mathematically, this can be expressed as:

$A \cap B = \varnothing$

Where:

- $A \cap B$ represents the intersection of set A and set B.
- $\varnothing$ (phi) represents the null set or empty set.

Key takeaway: If two sets are disjoint, they share no common elements.

Difference of Two Sets

Definition

The difference of two sets, denoted as $A - B$, represents the elements that are present in set A but not in set B. Similarly, $B - A$ represents the elements present in set B but not in set A.

Explanation

When we calculate $A - B$, we are essentially removing all the common elements (elements present in both A and B) from set A. The result is a set containing only those elements that are unique to A.

Example

Let's consider two sets:

1. $A - B$: To find $A - B$, we remove the common elements (3 and 4) from set A. $A - B$ = {1, 2}
2. $B - A$: To find $B - A$, we remove the common elements (3 and 4) from set B. $B - A$ = {5, 6}

Venn Diagram Representation

- $A - B$: This is represented by the region purely belonging to set A, excluding the intersection with set B.
- $B - A$: This is represented by the region purely belonging to set B, excluding the intersection with set A.

This operation helps in identifying elements unique to one set when compared to another.

Set Difference: A - B

Definition of A - B

The set difference $A - B$ (read as "A minus B") represents the region that purely belongs to set A, excluding any elements that are also in set B. It can also be denoted as $A \setminus B$.

Number of Elements in A - B

To find the number of elements in $A - B$, denoted as $n(A - B)$, we use the following formula:

$n(A - B) = n(A) - n(A \cap B)$

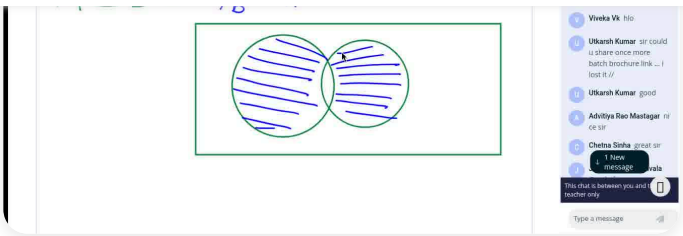
Where:

- $n(A - B)$ is the number of elements in the set $A - B$.
- $n(A)$ is the number of elements in set A.
- $n(A \cap B)$ is the number of elements in the intersection of set A and set B (elements common to both A and B).

Explanation

This formula states that from the total number of elements in set A, you must remove the elements that are common to both A and B (i.e., the intersection). This ensures that only the elements unique to A remain. While Venn diagrams are useful for visual understanding, this mathematical formula is crucial for calculations.

Symmetric Difference Between Two Sets



Definition and Notation

The symmetric difference between two sets, denoted as $A \Delta B$, represents the elements that are in either set A or set B, but not in their intersection. In simpler terms, it includes elements that belong purely to A or purely to B.

Formula

The symmetric difference can be expressed as the union of the difference between A and B, and the difference between B and A: $A \Delta B = (A - B) \cup (B - A)$ Alternatively, it can be written as: $A \Delta B = A / B \cup B / A$

Venn Diagram Representation

In a Venn diagram, for two overlapping sets A and B, the symmetric difference $A \Delta B$ corresponds to the regions that are exclusively part of A or exclusively part of B. This means the overlapping region (intersection) is excluded. It is the combination of the region purely belonging to A and the region purely belonging to B.

Complement of a Set

Definition

The complement of a set A, denoted as A^c or $A^{'}$, consists of all elements that are *not* in A but are within the universal set (or sample space, S).

Visual Representation

Imagine a universal set as a large rectangle. If set A is a circle inside this rectangle, then the complement of A (A^c) would be the area inside the rectangle but outside the circle.

Formula for Number of Elements

The number of elements in the complement of set A, denoted as $n(A^c)$, can be calculated using the following formula:

$n(A^c) = n(S) - n(A)$

- $n(S)$ is the total number of elements in the universal set (sample space).
- $n(A)$ is the number of elements in set A.

Definition of Complement

When discussing sets, the term "complement" refers to all elements that are *not* in a given set but are within the universal set under consideration.

Notations for Complement

There are several common ways to denote the complement of a set A:

- **A' (A dash):** This is a widely used notation.
- **A^c (A complement):** This notation explicitly uses the superscript 'c' for complement.
- **$\bar{A}$ (A bar):** This notation, using a bar symbol over the set name, also signifies the complement of set A.

All these notations mean the same thing: the complement of set A.

Properties of Complement of a Set

Complement of a Complement

If we take the complement of a set A, denoted as $A^{'}$, and then take the complement of $A^{'}$ again, we get the original set A.

$(A^{'})^{'} = A$

Union of a Set and its Complement

The union of a set A and its complement $A^{'}$ results in the universal set (U) or the complete sample space. This means combining all elements in A with all elements not in A covers everything in the universal set.

$A \cup A^{'} = U$

Intersection of a Set and its Complement

The intersection of a set A and its complement $A^{'}$ is always a null set ($\varnothing$). This is because A and $A^{'}$ contain no common elements by definition.

$A \cap A^{'} = \varnothing$

De Morgan's Laws

to simplify the complement of unions and intersections of sets.

There are two main De Morgan's Laws:

1. **Complement of a Union:** The complement of the union of two sets is equal to the intersection of their complements.
    - In terms of number of elements, if we denote the complement of a set A as $A^{'}$ (or $\bar{A}$ or A^{c}), then: $n(A \cup B)^{'} = n(A^{'} \cap B^{'})$
    - This means that elements not in $(A \cup B)$ are precisely those elements that are neither in A nor in B.
2. **Complement of an Intersection:** The complement of the intersection of two sets is equal to the union of their complements.
    - In terms of number of elements: $n(A \cap B)^{'} = n(A^{'} \cup B^{'})$
    - This means that elements not in $(A \cap B)$ are precisely those elements that are either not in A or not in B (or both).

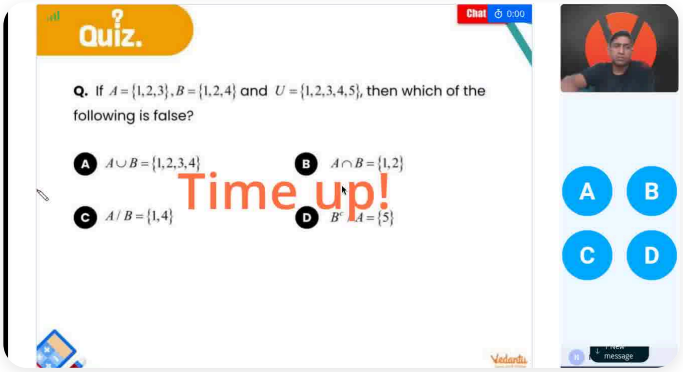
Notation for Complement

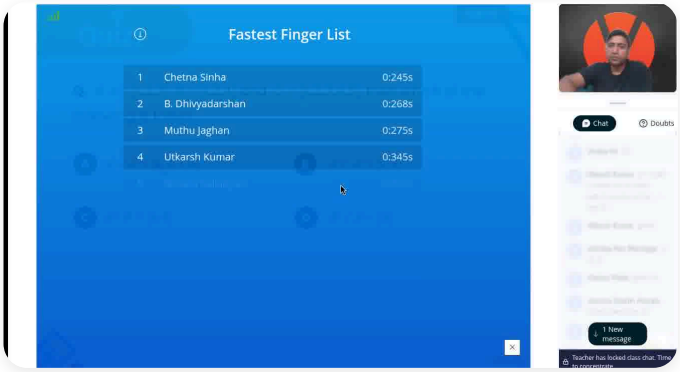
It is important to note that various symbols are used to represent the complement of a set. These include:

- Dash $(A^{'})$
- Bar $(\bar{A})$
- Superscript 'c' (A^{c})

All these symbols mean the same thing: the set of all elements in the universal set that are not in the given set.

Quiz: Set Operations





Notes



Question

Q. If A = {1, 2, 3}, B = {1, 2, 4} and U = {1, 2, 3, 4, 5}, then which of the following is false? A $A \cup B$ = {1, 2, 3, 4} B $A \cap B$ = {1, 2} C A/B = {1, 4} D $B^c \cap A$ = {5}

Solution Explanation

We need to identify the incorrect statement among the given options.

- **Option A: $A \cup B$ (Union of A and B)** The union of two sets A and B includes all elements that are in A, or in B, or in both. Given A = {1, 2, 3} and B = {1, 2, 4}, their union is $A \cup B$ = {1, 2, 3, 4}. This statement is **true**.
- **Option B: $A \cap B$ (Intersection of A and B)** The intersection of two sets A and B includes all elements that are common to both A and B. Given A = {1, 2, 3} and B = {1, 2, 4}, the common elements are 1 and 2. So, $A \cap B$ = {1, 2}. This statement is also **true**.
- **Option C: A/B (Difference of A and B)** The difference A/B (also written as $A - B$) includes all elements that are in A but not in B. From set A, we remove the elements that are common with B. The common elements are 1 and 2. So, A/B = {1, 2, 3} − {1, 2} = {3}. The option states A/B = {1, 4}, which is **false**. This is the incorrect statement.
- **Option D: $B^c \cap A$ (Complement of B intersected with A)** First, let's find the complement of B (B^c) with respect to the universal set U = {1, 2, 3, 4, 5}. Given B = {1, 2, 4}, $B^c = U - B$ = {1, 2, 3, 4, 5} − {1, 2, 4} = {3, 5}. Now, we find the intersection of B^c and A: $B^c \cap A$ = {3, 5} ∩ {1, 2, 3}. The common element is 3. So, $B^c \cap A$ = {3}. The option states $B^c \cap A$ = {5}, which is **false**.

  Self-correction during lecture: The lecturer initially states that $B^c \cap A$ = {5} is correct, but upon re-evaluation, it should be {3}. However, the question asks for the *false* statement. Let's re-check the lecturer's final conclusion. The lecturer states that option C is the wrong one, implying it's the incorrect answer. Let's re-evaluate option D based on the lecturer's final statement.

  Let's re-examine the lecturer's explanation for option D: "How about B complement? If set B is 1, 2, 4, what about B complement? I will say that will be 3, 5. And now from this, they want to remove A from 3, 5. If I have to remove this, I will get 5 only. So, this is also correct." The lecturer seems to have misspoken or made a slight error in the verbal explanation for option D,

Therefore, based on the provided options and calculations, option C is the false statement.

Final Answer

The final answer is $\boxed{C}$.

Difference and Symmetric Difference of Sets

Difference of Sets ($A - B$)

When we talk about the difference between two sets, A and B, denoted as $A - B$, we are referring to the elements that belong purely to set A. This means we take all elements in A and remove any elements that are also present in B (i.e., the intersection part).

Mathematically, $A - B = \{x \mid x \in A \text{ and } x \notin B\}$.

Symmetric Difference of Sets ($A \Delta B$)

The symmetric difference between two sets, A and B, denoted as $A \Delta B$, represents the elements that belong to either set A or set B, but not to their intersection. In simpler terms, it includes elements that are purely in A or purely in B.

It can be defined in two equivalent ways:

1. **Using Set Difference:** $A \Delta B = (A - B) \cup (B - A)$
    - This means we take the elements purely in A ($A - B$) and combine them with the elements purely in B ($B - A$).
2. **Using Union and Intersection:** $A \Delta B = (A \cup B) - (A \cap B)$
    - This means we take all elements in the union of A and B and remove the elements that are common to both (their intersection).

In essence, the symmetric difference focuses on those regions that are exclusively part of one set or the other, excluding any overlap.

Cardinality Formulas for Sets

Cardinality of Two Sets

For two sets, A and B, the number of elements in their union (A union B) is given by the formula:

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

Where:

- $n(B)$ represents the number of elements in set B.
- $n(A \cap B)$ represents the number of elements in the intersection of set A and set B.

Cardinality of Three Sets

For three sets, A, B, and C, the number of elements in their union (A union B union C) is given by the formula:

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$$

Where:

- $n(A \cup B \cup C)$ represents the number of elements in the union of sets A, B, and C.
- $n(A)$, $n(B)$, $n(C)$ represent the number of elements in sets A, B, and C, respectively.
- $n(A \cap B)$, $n(B \cap C)$, $n(C \cap A)$ represent the number of elements in the intersection of each pair of sets.
- $n(A \cap B \cap C)$ represents the number of elements in the intersection of all three sets.

Session Summary and Next Steps

Session Recap

This session focused on introducing the concept of sets. The pace was kept moderate to ensure everyone understood the foundational concepts. No application-based problems were covered in this introductory session.

Preparation for Next Session

The upcoming session will be heavily focused on calculations and practice questions. Students are advised to thoroughly revise the concepts covered in this session to be well-prepared for the next one.

Batch Information and Telegram Channel

Telegram Channel

Students are encouraged to join the batch's Telegram channel for important updates and discussions. The link to the Telegram channel has been shared with the students.

Batch Brochure

The batch brochure, containing details about the course, has also been shared with the students.

Session Recap and Leaderboard

Notes



Leaderboard

This section acknowledges the top performers in the session, highlighting their scores and ranks. The ranking system is based on the correctness of answers and bonus points for consistency, with speed being a key factor for overall rank.

- **Score & Rank Logic:**
  - Players receive scores based on the correctness of their answers.
  - Bonus points are awarded for consistently providing correct answers.
  - Ranking is determined by the speed at which players answer.

Top Performers:

- Aditya MH: 6
- Utkarsh Kumar: 6
- B. Dhivyadarshan: 6
- Shivani Kaliappan: 6
- Jayaramakreishnan Sr.: 6
- Jessica Sharlin Aluvala: 6
- Virgil: 6
- N.sarvaruban: 6
- Viveka Vk: 5
- Siddhartha: 5
- Dhanushree MR: 5

Important Announcements:

- **Telegram Channel:** AM Amit sir Kumar singh shared a Telegram link: `https://t.me/+YuhHWgBxGncwZmM1`

Next Steps:

The next session will focus on applications of the concepts covered. Students are encouraged to revise the current session's material thoroughly and come prepared.

Notes